Sinyal ve Sistemler

Sunu 2

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Basic Operations on Signals

There are two variable parameters in general:

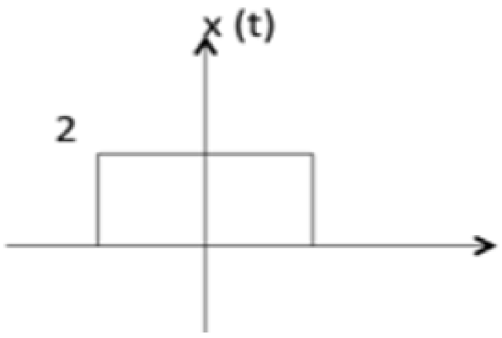
- Amplitude
- Time

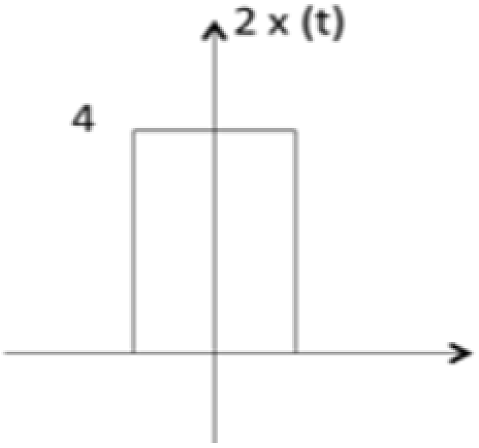
The following operation can be performed with amplitude:

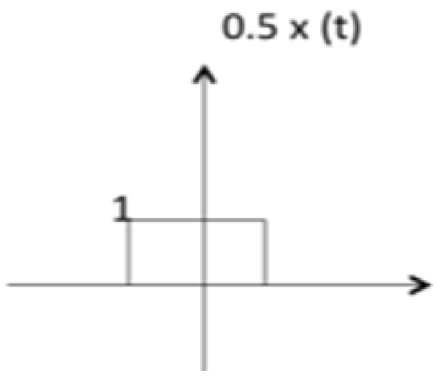
Amplitude Scaling

C x(t) is a amplitude scaled version of x(t) whose amplitude is scaled by a factor C.

Exp: What is the 2x(t) and 0.5x(t)



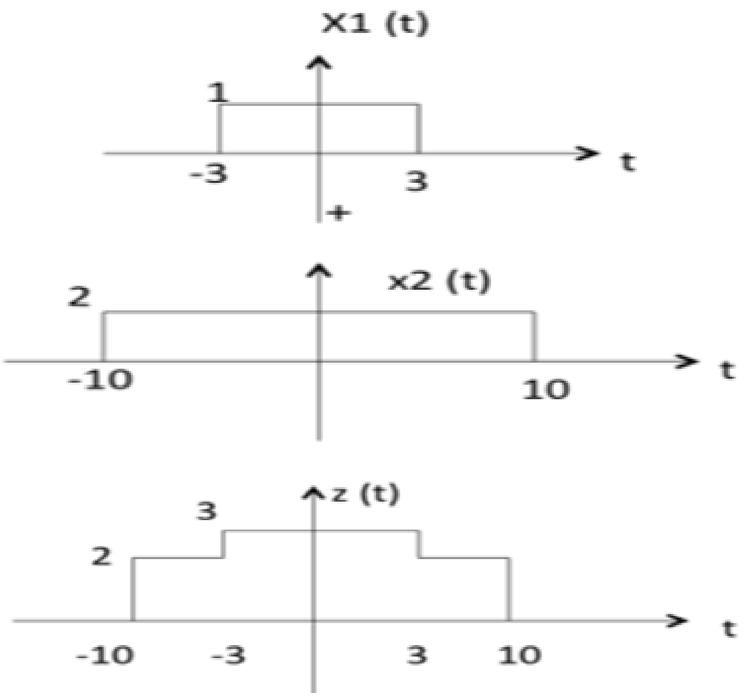




Basic Operations on Signals

Addition

Addition of two signals is nothing but addition of their corresponding amplitudes. This can be best explained by using the following example:



As seen from the previous diagram,

$-10 < t < -3$ amplitude of $z(t) = x1(t) + x2(t) = 0 + 2 = 2$

$-3 < t < 3$ amplitude of $z(t) = x1(t) + x2(t) = 1 + 2 = 3$

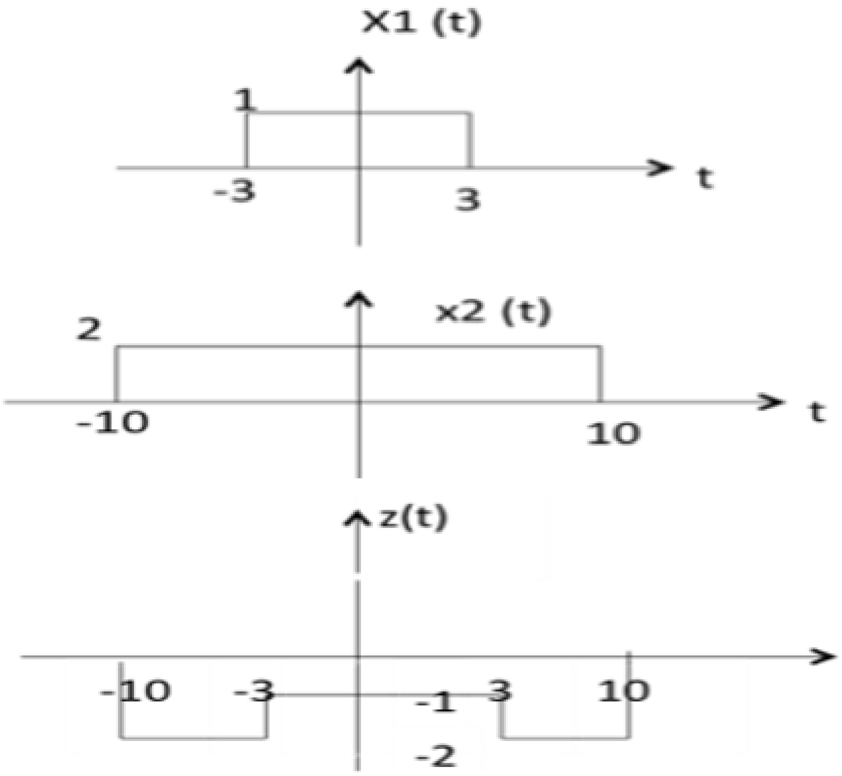
$3 < t < 10$ amplitude of $z(t) = x1(t) + x2(t) = 0 + 2 = 2$

Basic Operations on Signals

Subtraction

subtraction of two signals is nothing but subtraction of their corresponding amplitudes.

This can be best explained by the following example:



As seen from the diagram above,

$-10 < t < -3$ amplitude of $z(t) = x1(t) - x2(t) = 0 - 2 = -2$

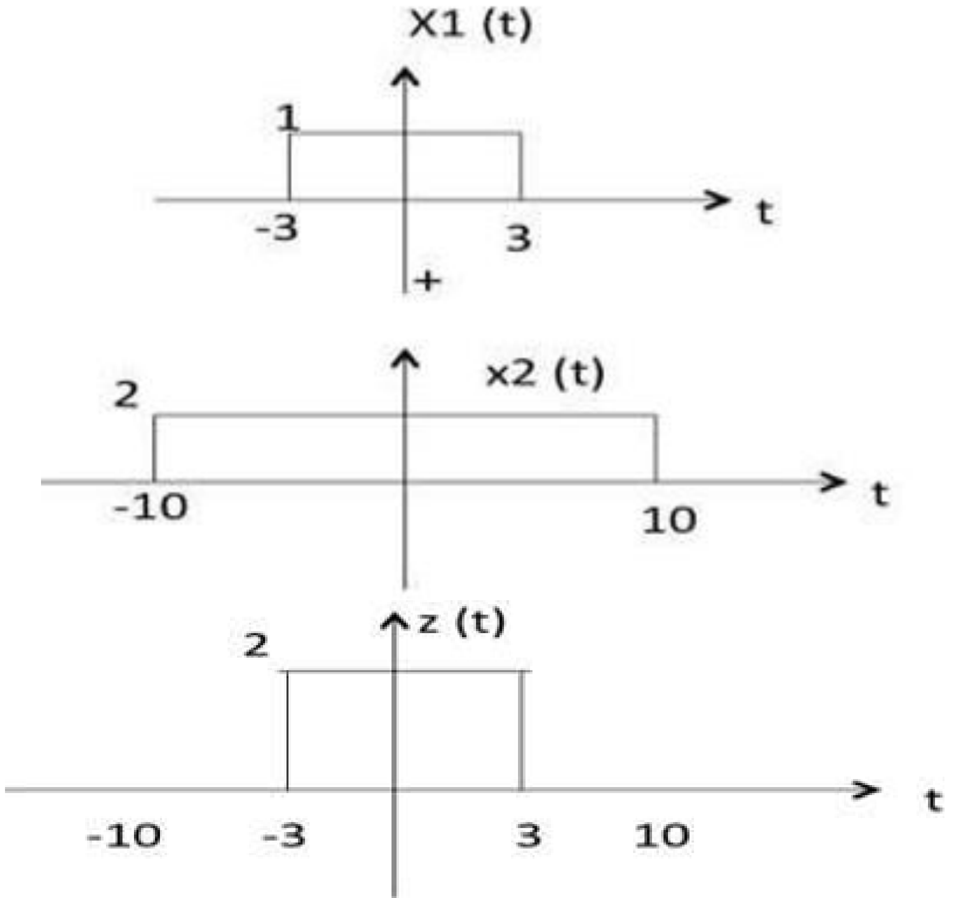
$-3 < t < 3$ amplitude of $z(t) = x1(t) - x2(t) = 1 - 2 = -1$

$3 < t < 10$ amplitude of $z(t) = x1(t) - x2(t) = 0 - 2 = -2$

Basic Operations on Signals

Multiplication

Multiplication of two signals is nothing but multiplication of their corresponding amplitudes. This can be best explained by the following example:



As seen from the diagram above,

$-10 < t < -3$ amplitude of $z(t) = x1(t) \times x2(t) = 0 \times 2 = 0$

$-3 < t < 3$ amplitude of $z(t) = x1(t) - x2(t) = 1 \times 2 = 2$

$3 < t < 10$ amplitude of $z(t) = x1(t) - x2(t) = 0 \times 2 = 0$

Basic Operations on Signals

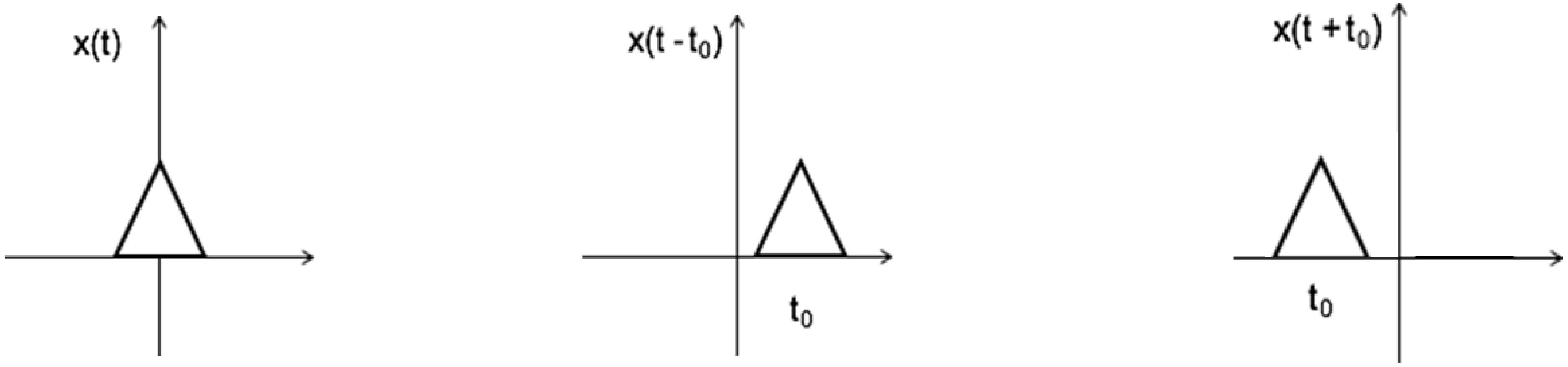
The following operations can be performed with time:

Time Shifting

x(t ±t0) is time shifted version of the signal x(t).

x (t - t0) →positive shift

x (t + t0) →negative shift



EXAMPLE 1.4 TIME SHIFTING Figure 1.23(a) shows a rectangular pulse $x(t)$ of unit amplitude and unit duration. Find $y(t) = x(t - 2)$.

Solution: In this example, the time shift t_0 equals 2 time units. Hence, by shifting $x(t)$ to the right by 2 time units, we get the rectangular pulse $y(t)$ shown in the figure. The pulse $y(t)$ has exactly the same shape as the original pulse $x(t)$; it is merely shifted along the time axis. ■

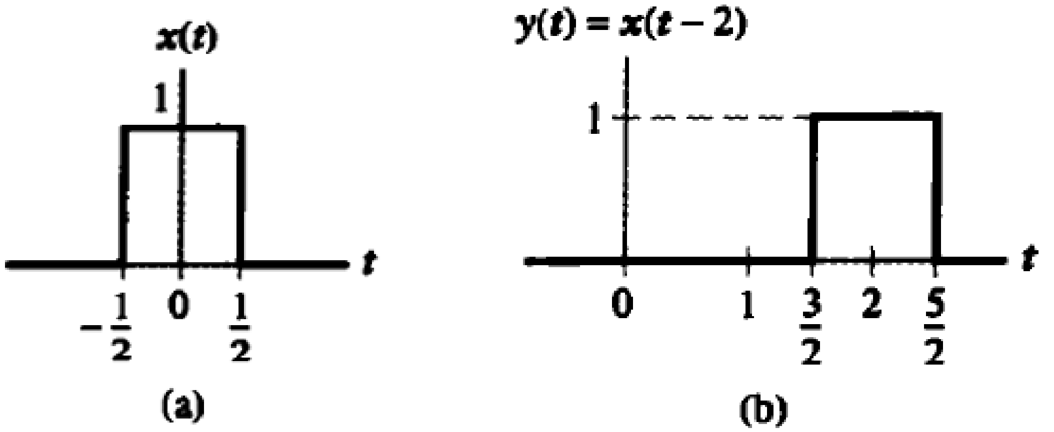
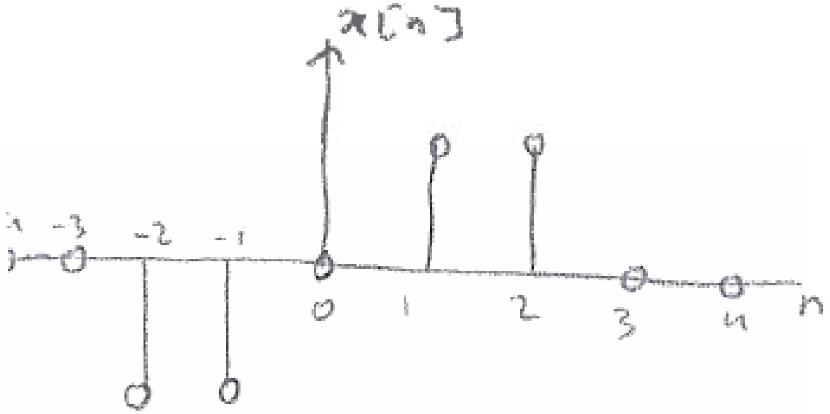
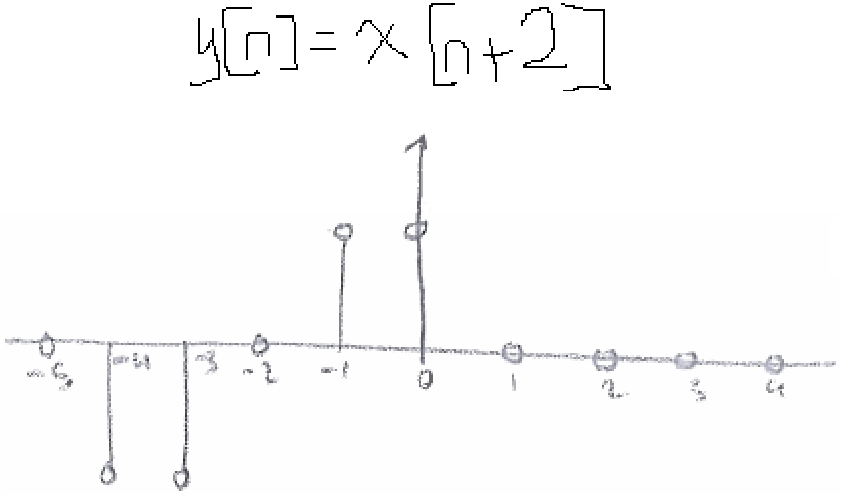


FIGURE 1.23 Time-shifting operation: (a) continuous-time signal in the form of a rectangular pulse of amplitude 1 and duration 1, symmetric about the origin; and (b) time-shifted version of $x(t)$ by 2 time units.

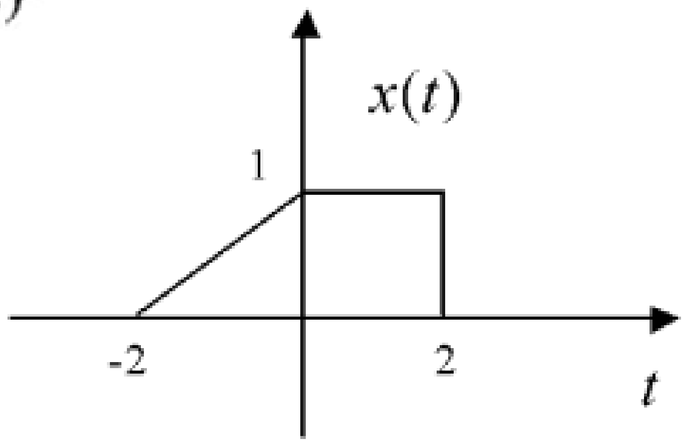
Exp what is the y[n]= x[n+2]





Time Shifting

Exp: $y(t) = x(t+2)$, $y(t) = x(t-2)$



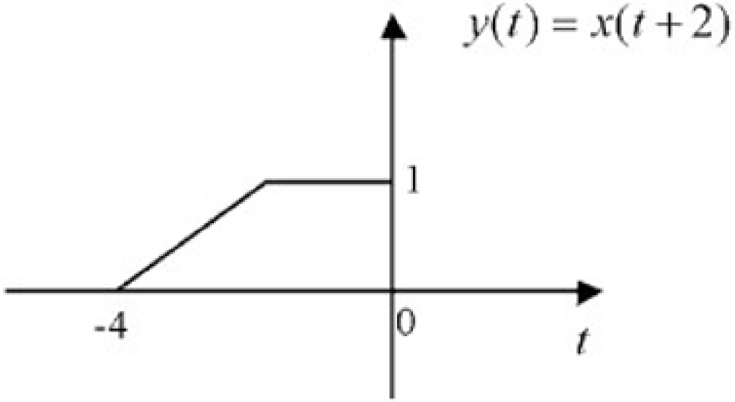


FIGURE 1.11 Graph of time-advanced signal $y(t) = x(t+2)$.

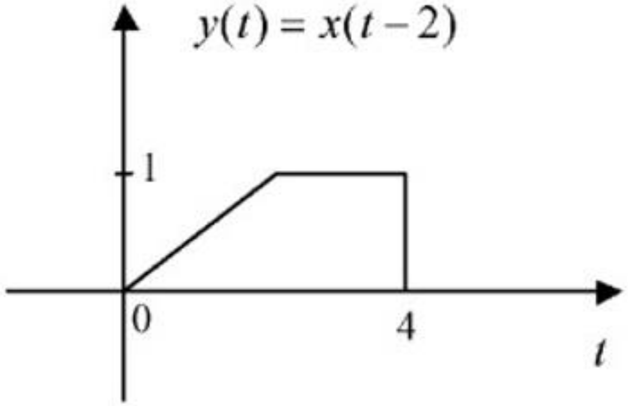


FIGURE 1.12 Graph of time-delayed signal $y(t) = x(t-2)$.

Time Shift

Exp: $y[n] = x[n+2]$

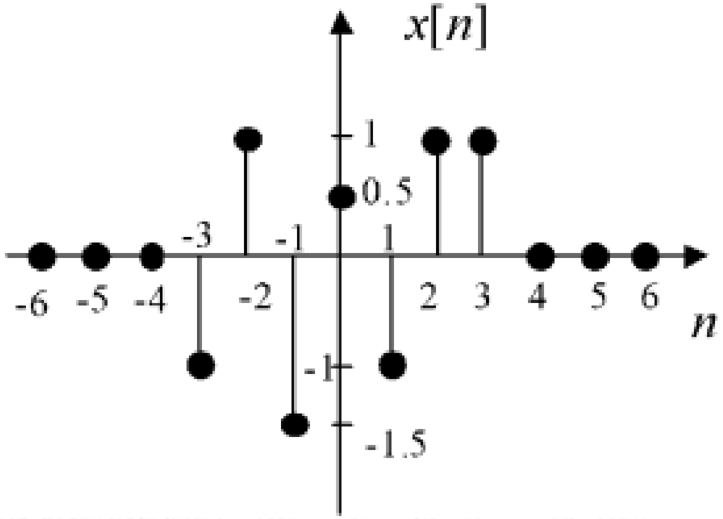


FIGURE 1.5 Graph of discrete-time signal $x[n]$.

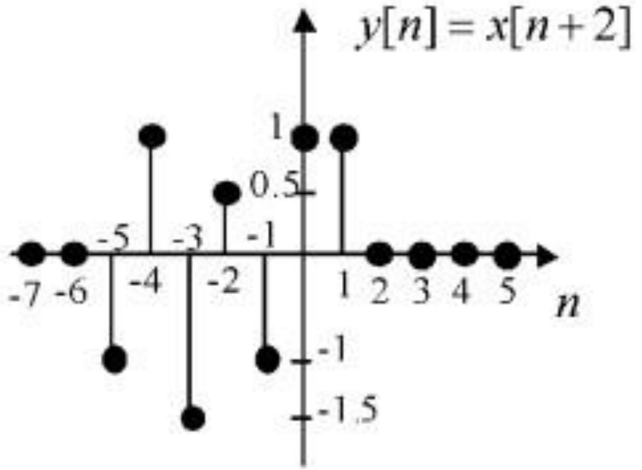


FIGURE 1.13 Graph of time-advanced signal $y[n] = x[n + 2]$.

Basic Operations on Signals

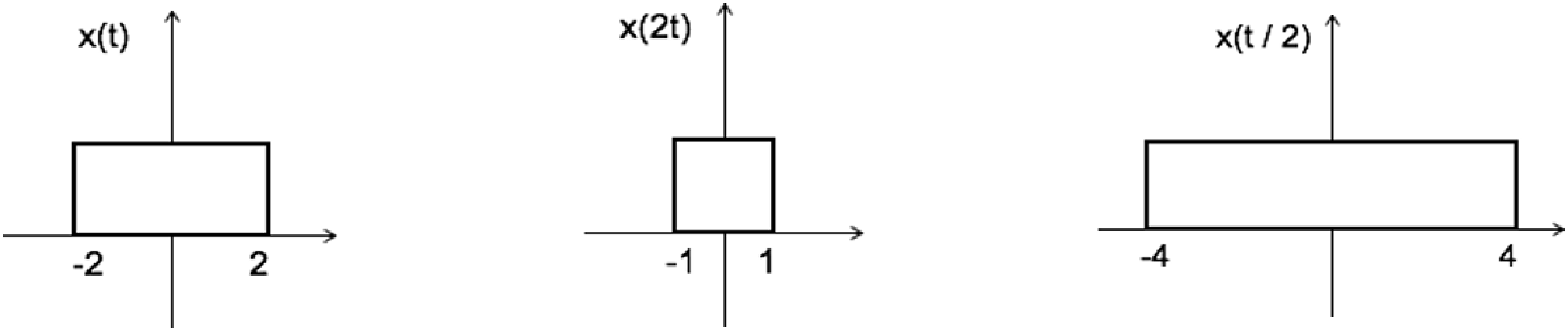
Time Scaling

x(At) is time scaled version of the signal x(t). where A is always positive.

| A | > 1 → Compression of the signal

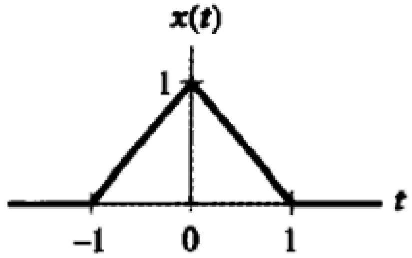
| A | < 1 → Expansion of the signal

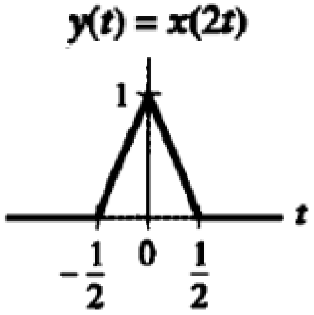
Exp: Find x(2t) and x(t/2)

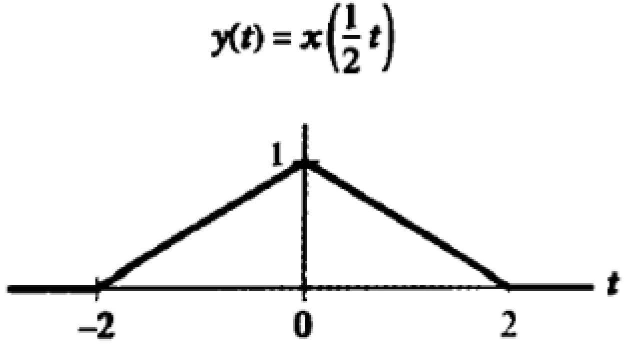


Note: u(at) = u(t) time scaling is not applicable for unit step function.

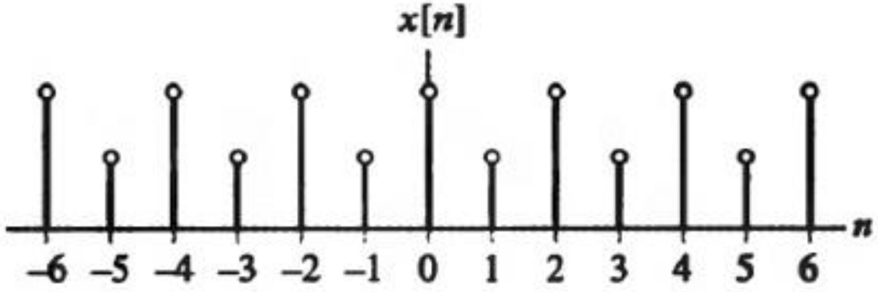
- Exp: $y(t) = x(2t)$, $y(t) = x\left(\frac{1}{2}t\right)$

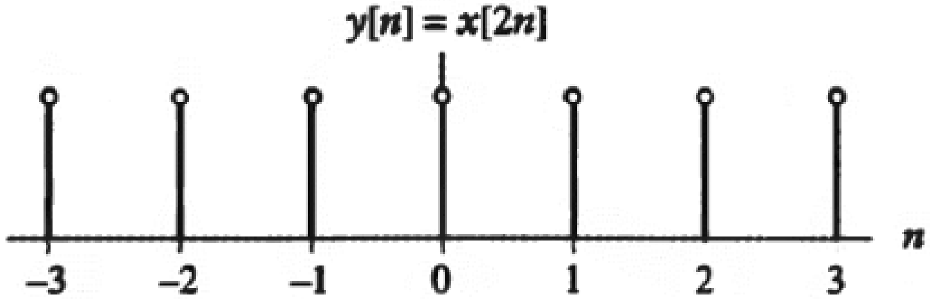






- Exp: $y[n] = x[2n]$





Exp:

Find: y(t)=x(0.5t) , y(t)=x(2t)

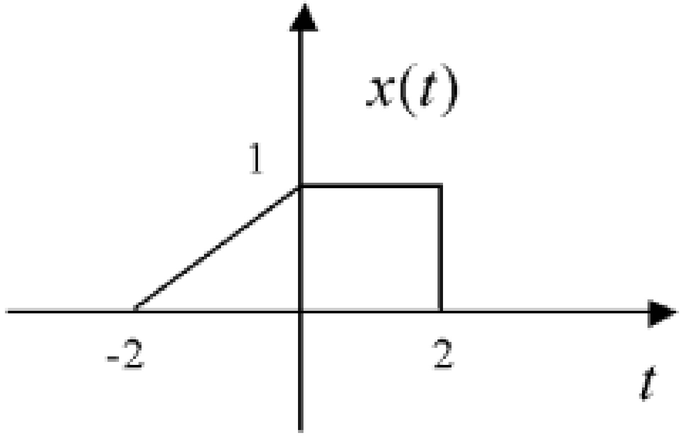


FIGURE 1.4 Graph of continuous time signal $x(t)$.

Example 1.3: Case $\alpha = \frac{1}{2}$ shown in Figure 1.6.

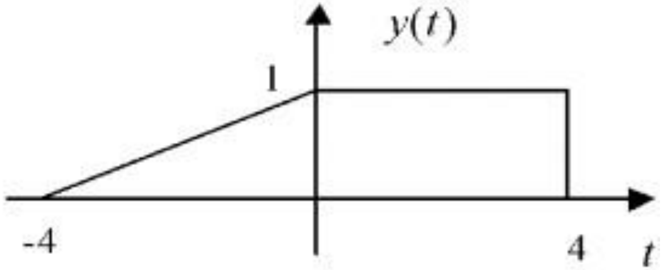


FIGURE 1.6 Graph of expanded signal $y(t) = x(0.5t)$.

Case $0 < \alpha < 1$: The signal $x(t)$ is *slowed down* or *expanded* in time. Think of a tape recording played back at a slower speed than the nominal speed.

Example 1.4: Case $\alpha = 2$ shown in Figure 1.7.

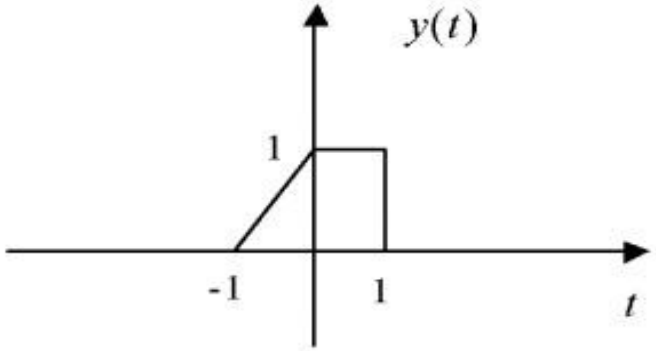


FIGURE 1.7 Graph of compressed signal $y(t) = x(2t)$.

Case $\alpha > 1$: The signal $x(t)$ is *sped up* or *compressed* in time. Think of a tape recording played back at twice the nominal speed.

Exp:

Find: y[n]=x[2n]

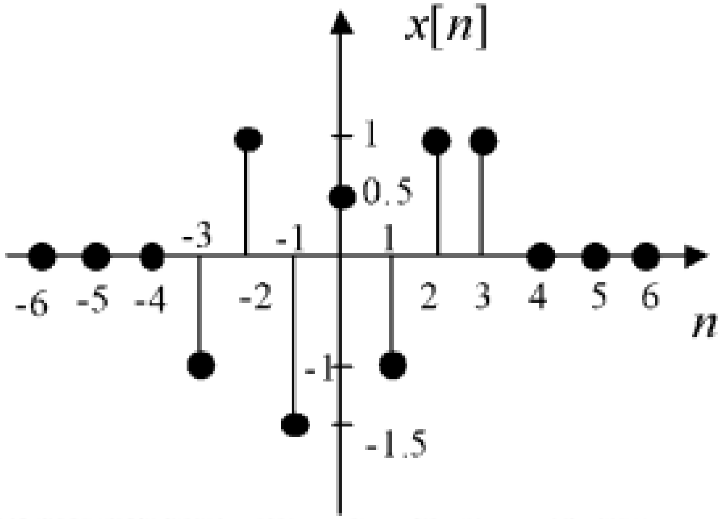


FIGURE 1.5 Graph of discrete-time signal $x[n]$.

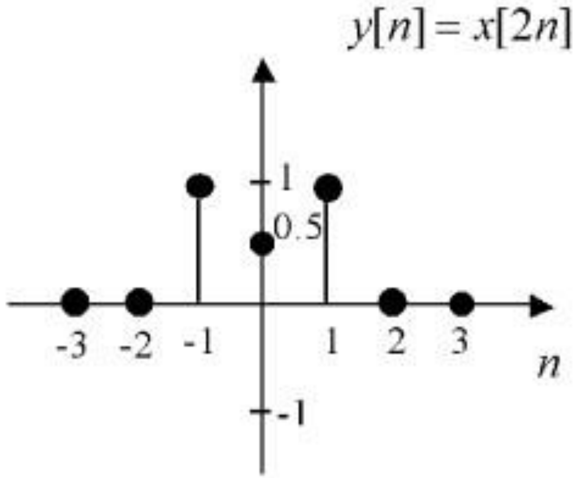


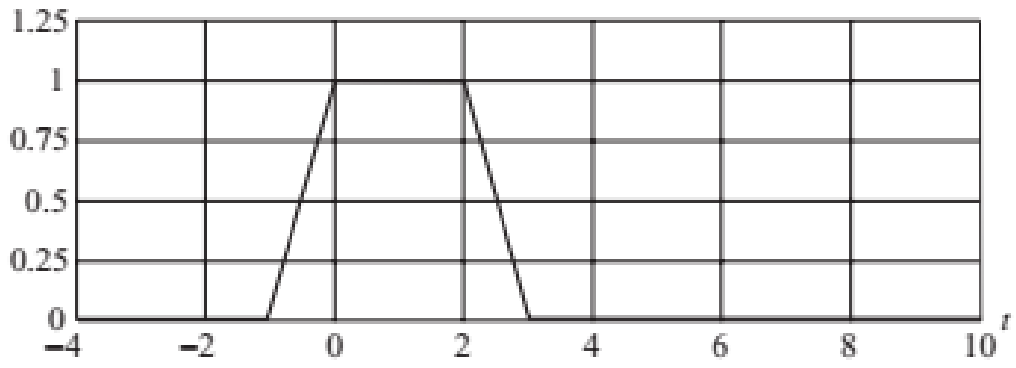
FIGURE 1.8 Graph of compressed signal $y[n] = x[2n]$.

Example 1.16

Consider a CT signal $x(t)$ defined as follows:

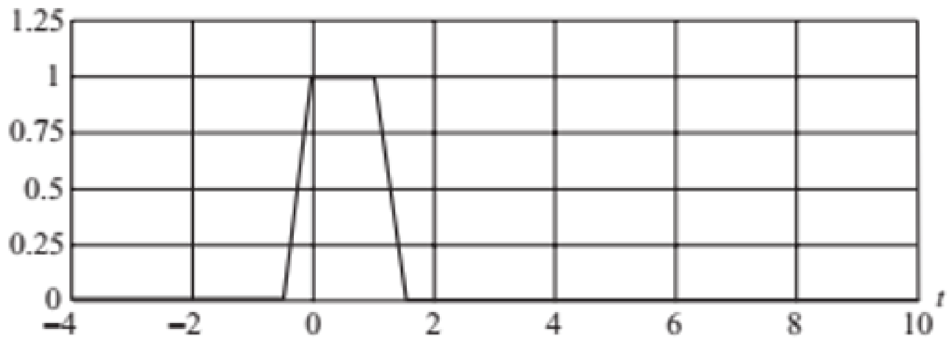
$$x(t) = \begin{cases} t+1 & -1 \le t \le 0 \\ 1 & 0 \le t \le 2 \\ -t+3 & 2 \le t \le 3 \\ 0 & \text{elsewhere,} \end{cases} \tag{1.53}$$

as plotted in Fig. 1.25(a). Determine the expressions for the time-scaled signals $x(2t)$ and $x(t/2)$. Sketch the two signals.

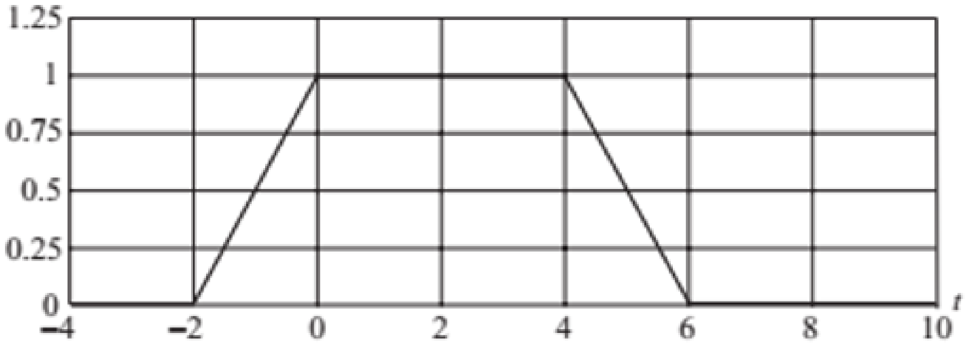


(a)

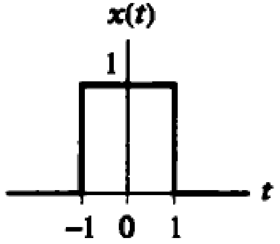
y(t)=x(2t)



y(t)=x(t/2)



EXAMPLE 1.5 PRECEDENCE RULE FOR CONTINUOUS-TIME SIGNAL Consider the rectangular pulse $x(t)$ of unit amplitude and a duration of 2 time units, depicted in Fig. 1.24(a). Find $y(t) = x(2t + 3)$.



Solution: In this case, the first time shift and then time scaling have to be done

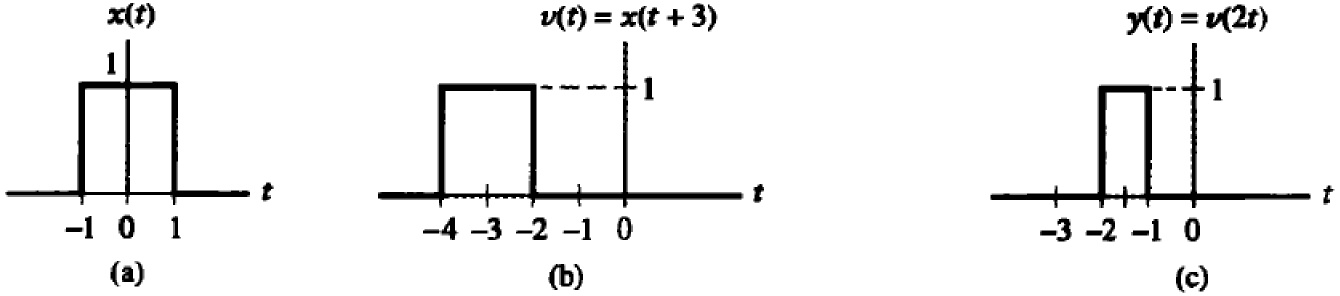


FIGURE 1.24 The proper order in which the operations of time scaling and time shifting should be applied in the case of the continuous-time signal of Example 1.5. (a) Rectangular pulse $x(t)$ of amplitude 1.0 and duration 2.0, symmetric about the origin. (b) Intermediate pulse $v(t)$, representing a time-shifted version of $x(t)$. (c) Desired signal $y(t)$, resulting from the compression of $v(t)$ by a factor of 2.

- Otherwise, the solution will not be correct like this:

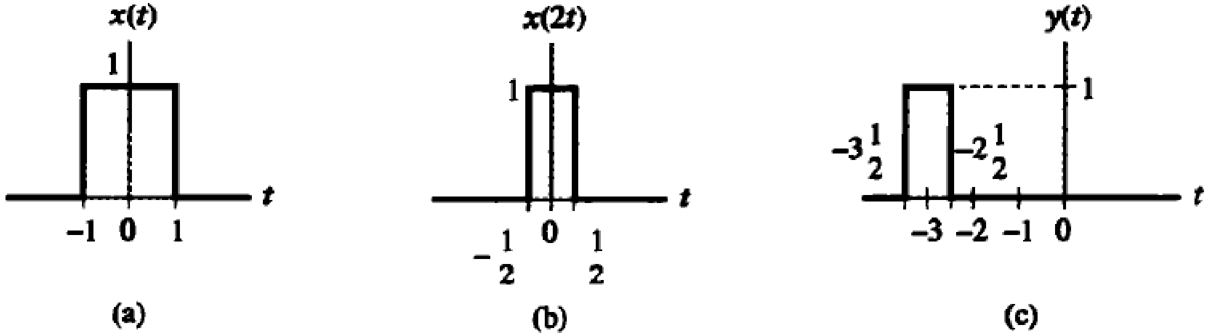
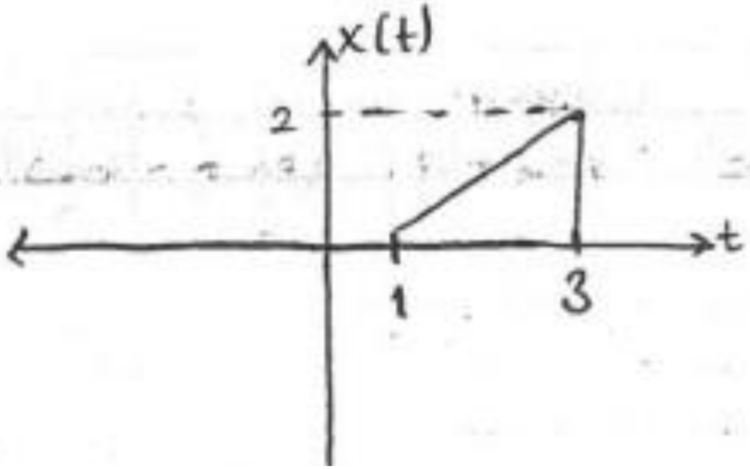


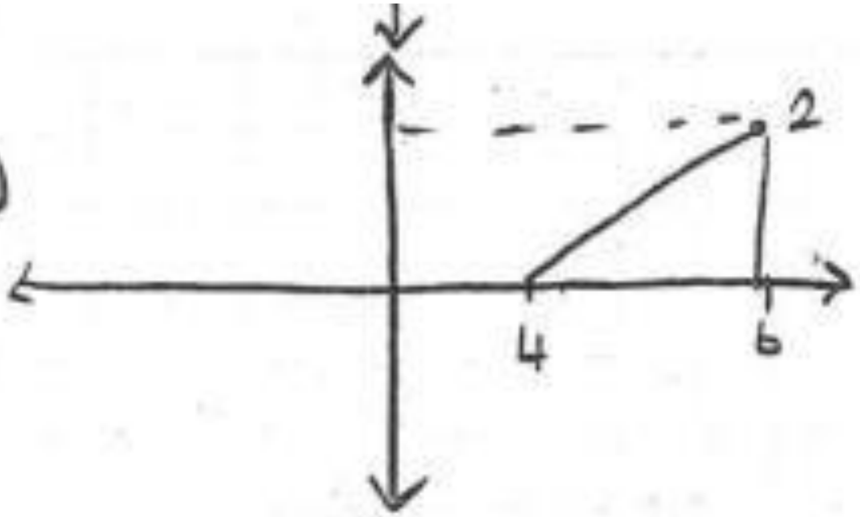
FIGURE 1.25 The incorrect way of applying the precedence rule. (a) Signal $x(t)$. (b) Time-scaled signal $v(t) = x(2t)$. (c) Signal $y(t)$ obtained by shifting $v(t) = x(2t)$ by 3 time units, which yields $y(t) = x(2(t + 3))$.

Örnek:

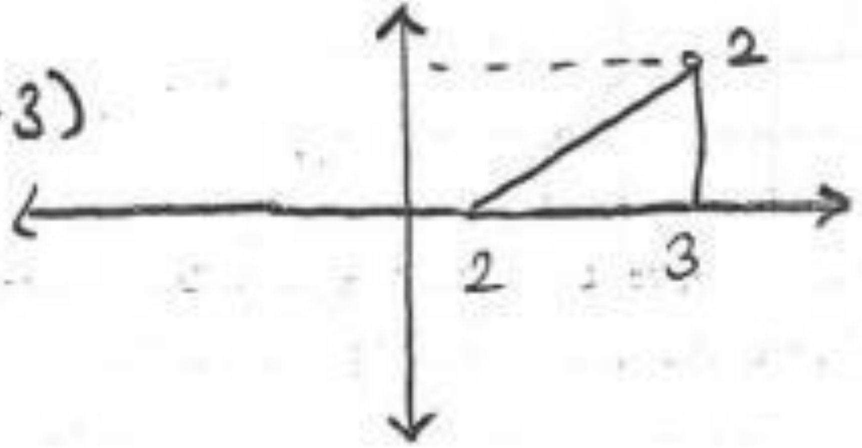


$x(t) \longrightarrow x(2t-3)$

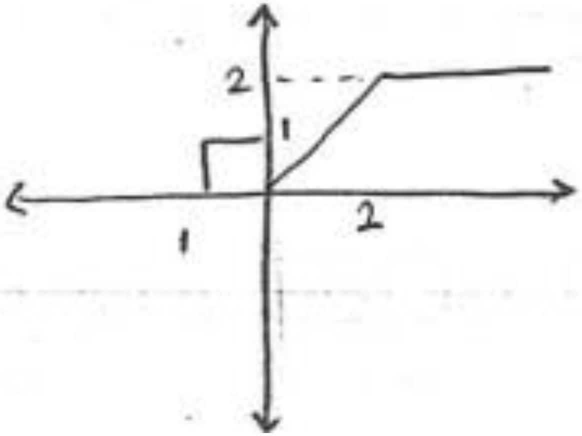
$x(t-3)$



$x(2t-3)$

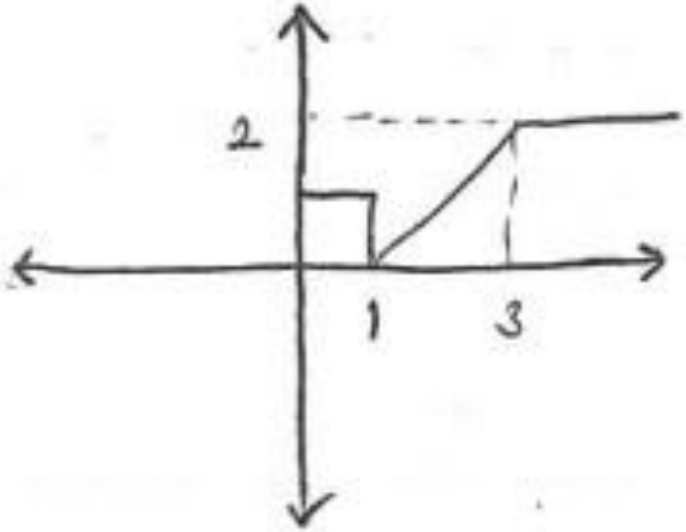


Örnek:

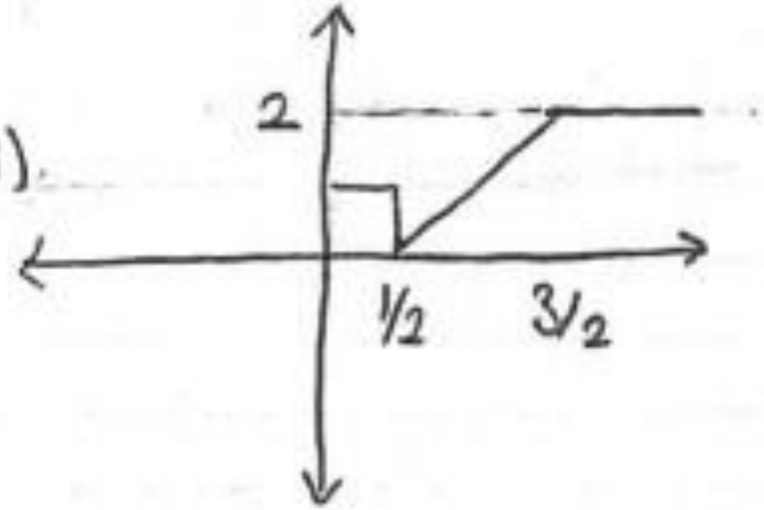


$x(t) \longrightarrow x(2t-1)$

$x(t-1)$



$x(2t-1)$

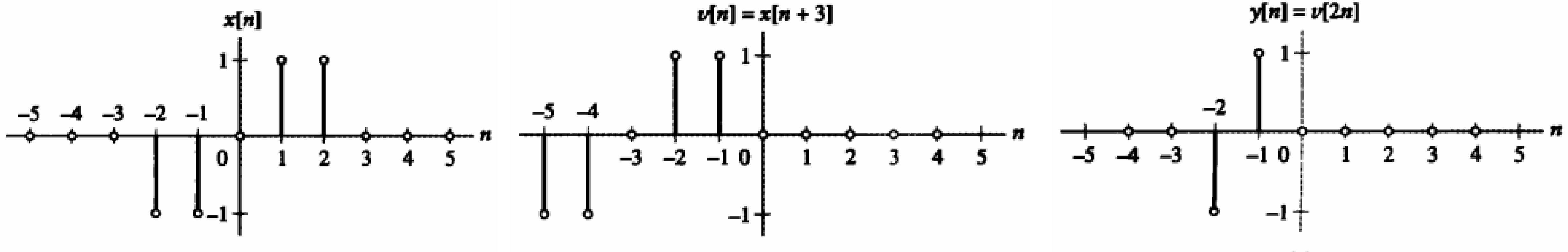


EXAMPLE 1.6 PRECEDENCE RULE FOR DISCRETE-TIME SIGNAL A discrete-time signal is defined by

$$x[n] = \begin{cases} 1, & n = 1, 2 \\ -1, & n = -1, -2 \\ 0, & n = 0 \text{ and } |n| > 2 \end{cases}.$$

Find $y[n] = x[2n + 3]$.

Solution:

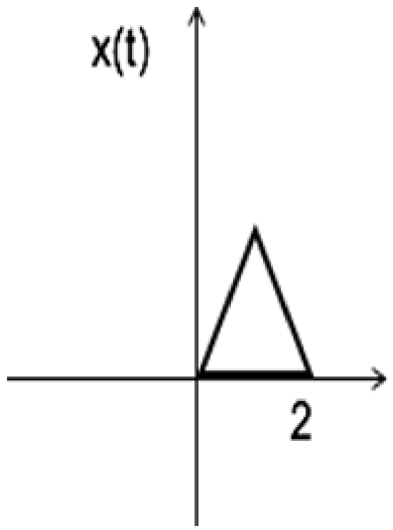


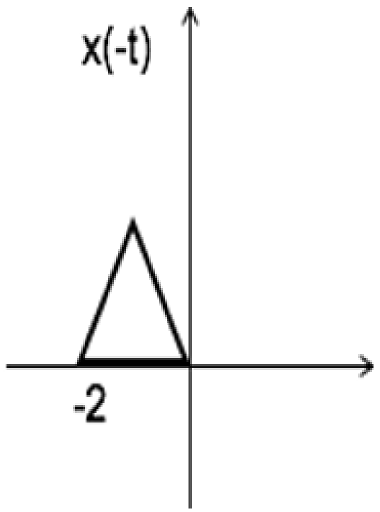
(a) (b) (c)

Basic Operations on Signals

Time Reversal

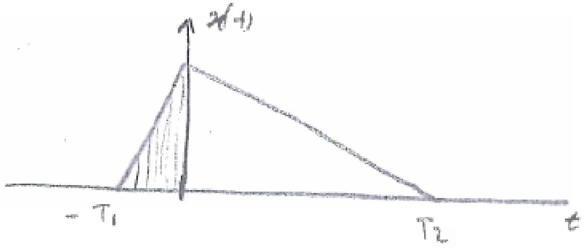
x(-t) is the time reversal of the signal x(t).

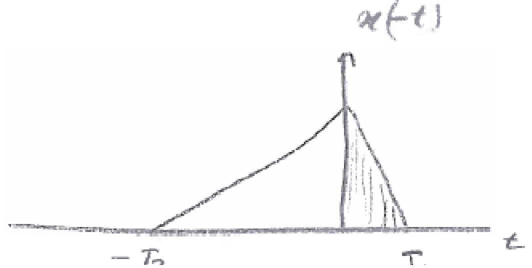




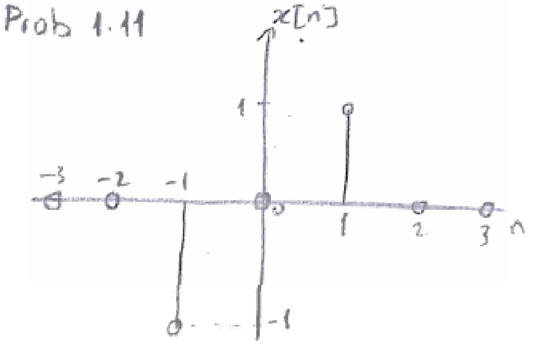
Reflection (Yansıma)

$y(t) = x(-t)$

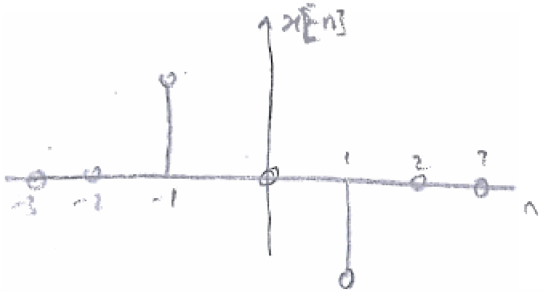


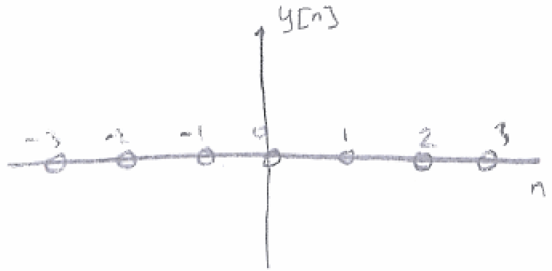


Prob 1.11

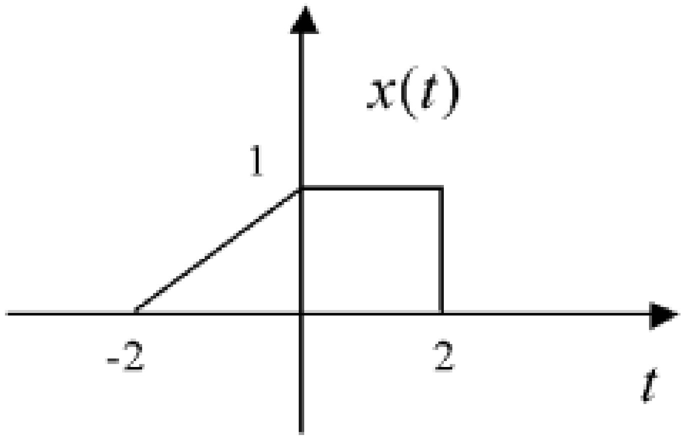


$y[n] = x[n] + x[-n]$





Time Reversal



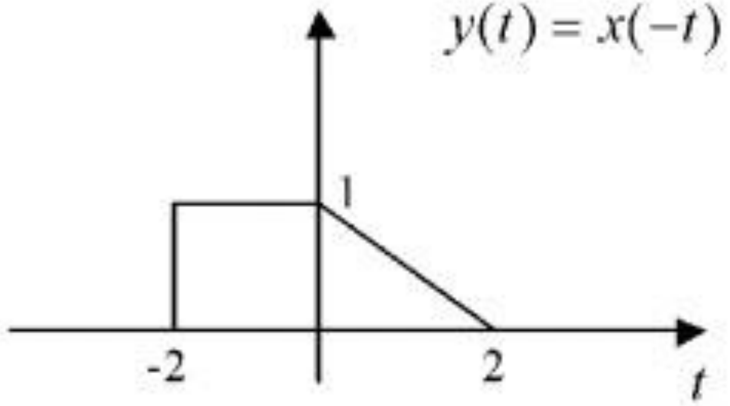


FIGURE 1.9 Graph of time-reversed signal $y(t) = x(-t)$.

Time Reversal

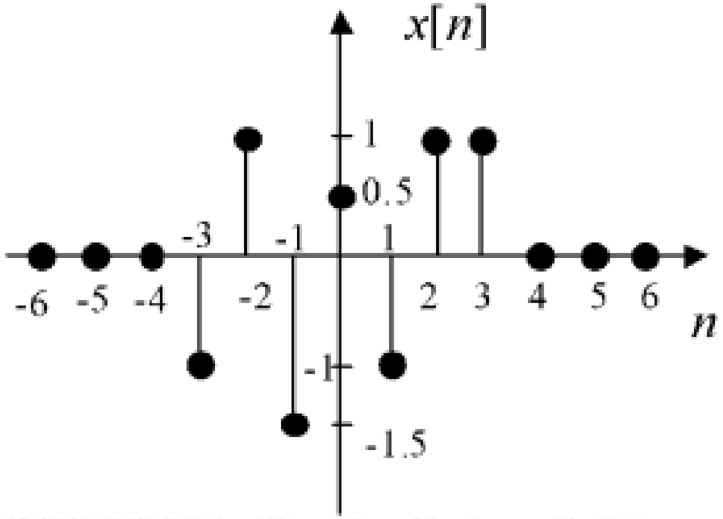


FIGURE 1.5 Graph of discrete-time signal $x[n]$.

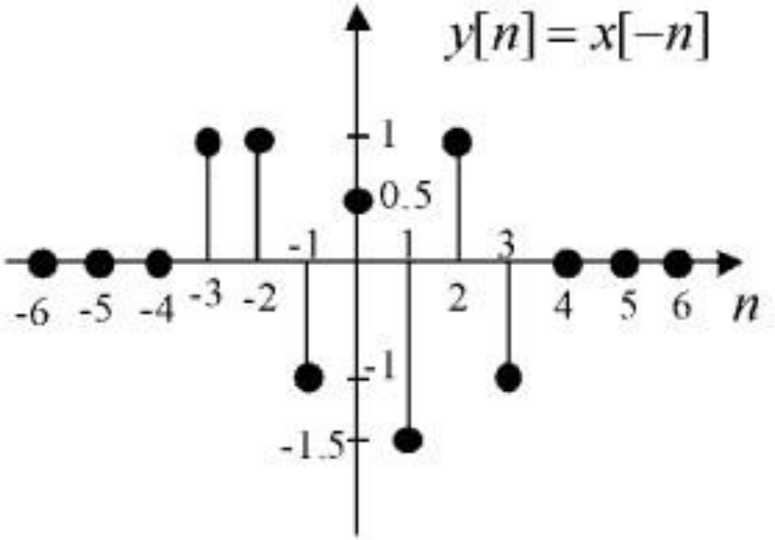


FIGURE 1.10 Graph of time-reversed signal $y[n] = x[-n]$.

▶ **Problem 1.14** A triangular pulse signal $x(t)$ is depicted in Fig. 1.26. Sketch each of the following signals derived from $x(t)$:

(a) $x(3t)$

(b) $x(3t + 2)$

(c) $x(-2t - 1)$

(d) $x(2(t + 2))$

(e) $x(2(t - 2))$

(f) $x(3t) + x(3t + 2)$

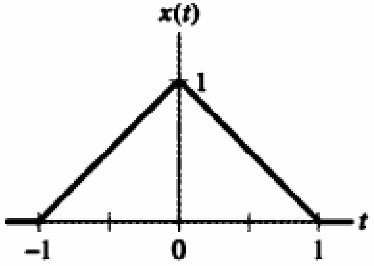


FIGURE 1.26 Triangular pulse for Problem 1.14.